

Using u-Substitution in Indefinite and Definite Integrals

Answer Key

AP Calculus AB / BC Integration Techniques

Quick Answers

1. $\frac{(x^2+5)^4}{4}$

2. $\frac{\sin(x^2)}{2}$

3. $\frac{\log(x)^3}{3}$

4. $-\frac{1}{2} + \frac{e^4}{2}$

5. $\frac{\sin^4(x)}{4}$

6. $\log(5)$

7. $\frac{2(x^3+1)^{\frac{3}{2}}}{9}$

8. $\frac{1}{2}, \frac{1}{2}$

1. $\int 2x(x^2+5)^3 dx.$

Step 1. Look for an inner function whose derivative also appears. The inner function is $x^2 + 5$, so let $u = x^2 + 5$; then $du = 2x dx$, which is exactly the factor sitting outside the bracket.

Step 2. The integral becomes $\int u^3 du = \frac{u^4}{4}$, and back-substituting returns the answer in

x .

$$\frac{(x^2+5)^4}{4} + C$$

Check by differentiating: $\frac{4(x^2+5)^3 \cdot 2x}{4} = 2x(x^2+5)^3$, the original integrand.

2. $\int x \cos(x^2) dx.$

Step 1. Let $u = x^2$, so $du = 2x dx$. The integrand offers only $x dx$, which is $\frac{1}{2} du$ — a constant short, not a function short.

Step 2. Part (b): a constant may be moved across the integral sign because $\int kf = k \int f$; a variable may not, since x changes as the integration variable moves. So $\int x \cos(x^2) dx =$

$$\frac{1}{2} \int \cos u du = \frac{1}{2} \sin u.$$

Check: $\frac{1}{2} \cos(x^2) \cdot 2x = x \cos(x^2)$.

$$\frac{\sin(x^2)}{2} + C$$

3. $\int \frac{(\ln x)^2}{x} dx.$

Step 1. Part (a): $\ln x$ is the inner function and $\frac{1}{x}$ is its derivative, so the integrand is already in the form $f(u) du$.

Step 2. With $u = \ln x$ and $du = \frac{1}{x} dx$, the integral is $\int u^2 du = \frac{u^3}{3}$.

Check: $\frac{3(\ln x)^2}{3} \cdot \frac{1}{x} = \frac{(\ln x)^2}{x}$.

$$\frac{(\ln x)^3}{3} + C$$

4. $\int_0^2 x e^{x^2} dx.$

Step 1. Let $u = x^2$, so $du = 2x dx$ and $x dx = \frac{1}{2}du$. Because the integral is definite, convert the limits too: $x = 0$ gives $u = 0$, and $x = 2$ gives $u = 4$.

Step 2. The integral becomes $\frac{1}{2} \int_0^4 e^u du = \frac{1}{2} [e^u]_0^4 = \frac{e^4 - 1}{2}.$

$$\frac{e^4 - 1}{2}$$

Why convert the limits: once the bounds are in u , no back-substitution is needed. Evaluating $[\frac{1}{2}e^u]$ at the *old* bounds 0 and 2 is the single most common error on this problem type.

5. $\int \sin^3 x \cos x dx.$

Step 1. Part (a): with $u = \sin x$, $du = \cos x dx$; with $u = \cos x$, $du = -\sin x dx$. Test each against what the integrand actually supplies before committing.

Step 2. Part (b): $u = \sin x$ consumes the whole integrand, since $\sin^3 x \cos x dx = u^3 du$. The other choice absorbs only one $\sin x$ and strands $\sin^2 x \cos x$ behind, which cannot be written in u alone.

Step 3. $\int u^3 du = \frac{u^4}{4}$, then back-substitute.

$$\frac{\sin^4 x}{4} + C$$

Check: $\frac{4\sin^3 x \cdot \cos x}{4} = \sin^3 x \cos x.$

6. $\int_1^3 \frac{2x}{x^2 + 1} dx.$

Step 1. Let $u = x^2 + 1$, so $du = 2x dx$ — the numerator exactly. New limits: $x = 1$ gives $u = 2$, and $x = 3$ gives $u = 10$.

Step 2. $\int_2^{10} \frac{du}{u} = [\ln |u|]_2^{10} = \ln 10 - \ln 2.$

Step 3. Part (c): a difference of logarithms is the logarithm of the quotient, and $10 \div 2 = 5$.

$$\ln 5$$

Note on absolute values: on the new interval u runs from 2 to 10, so $u > 0$ throughout and the absolute value can be dropped without comment.

7. $\int x^2 \sqrt{x^3 + 1} dx.$

Step 1. Let $u = x^3 + 1$, so $du = 3x^2 dx$ and $x^2 dx = \frac{1}{3}du$. The integral carries a factor of one third.

Step 2. Part (b): write the root as a power — $\frac{1}{3} \int u^{1/2} du$ — so the power rule applies.

Step 3. $\frac{1}{3} \cdot \frac{u^{3/2}}{3/2} = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} = \frac{2}{9} u^{3/2}.$

$$\frac{2(x^3 + 1)^{3/2}}{9} + C$$

Check: $\frac{2}{9} \cdot \frac{3}{2} (x^3 + 1)^{1/2} \cdot 3x^2 = x^2 \sqrt{x^3 + 1}.$

8. Challenge: $\int_0^\infty x e^{-x^2} dx$.

Step 1. Part (a): infinity is not a value the Fundamental Theorem can be evaluated at, so the integral is *defined* as $\lim_{b \rightarrow \infty} \int_0^b x e^{-x^2} dx$. Everything else is an ordinary substitution.

Step 2. Let $u = -x^2$, so $du = -2x dx$ and $x dx = -\frac{1}{2}du$. The antiderivative is $-\frac{1}{2}e^{-x^2}$, and on $[0, b]$ it evaluates to $-\frac{1}{2}e^{-b^2} + \frac{1}{2}$.

Step 3. Part (b): as $b \rightarrow \infty$, $e^{-b^2} \rightarrow 0$, so the limit is the constant term that survives.

$$\boxed{\frac{1}{2}}$$

Step 4. Part (c): the limit exists and is finite, so the improper integral converges, and its value is that limit.

$$\boxed{\frac{1}{2}}$$

Why the substitution still governs: the limit only decides convergence. The number itself came from the antiderivative u -substitution produced — and note that $u = x^2$ works just as well, giving $\frac{1}{2} (1 - e^{-b^2})$, the same expression rearranged.